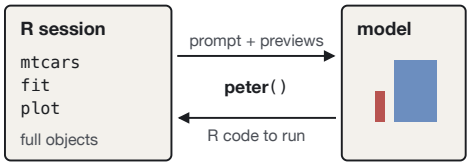
Agents in your R session with gptr :: CHEATSHEET

`peter()` runs language-model agents on the objects already in your R session. Named for Peter Wason, whose dual-process research inspires its System 1 and System 2 roles, and Peter Naur, of Backus-Naur form: sessions become readable, runnable documents.

How it works



Objects stay in your R session and the agent's R code runs there on the full data. An agent sees the prompt, bounded previews and tool output; a decision model is sent each input.

gptr_describe(mtcars)
What a model is shown of an object.

Set up

pak::pak("Broccolito/gptr@main")
library(gptr)
Install gptr 1.0 from GitHub (R 4.2 or later).

gptr_env("~/keys/.env")
Load provider keys from a private .env file; values are registered for redaction.

gptr_login("openrouter")
Sign in interactively: OAuth for OpenRouter, masked key entry for other providers.

gptr_providers()
Which routes are ready: credential, status, default model, egress.

gptr_models("sonnet")
Search the offline model catalog by name, alias or provider.

gptr_config(model = opus, .scope = "user")
Save a default model. Scopes: "session", "project", "user".

gptr_init(".")
gptr_trust(".", trust = TRUE)
Create a .gptr/ project workspace, then trust the project.

gptr_permissions(allow = "r(fn:saveRDS)")
Add a permission rule for this R session; no arguments lists the rules.

fake = **gptr_fake_provider**(list("32 rows."))
A scripted offline model for examples and tests: no key, no network.

With no model configured, the first interactive `peter()` offers the Codex CLI, the Claude Code CLI or manual API or Ollama setup.

Ask Peter from R

Give `peter()` a prompt and the objects to work on; it returns a session. It works in scripts, loops and functions.

s = **peter**("Fit mpg on wt; save it as fit.", mtcars)
The prompt first, then the objects to work on.

s$text; s$value; s$turns
Final answer, the value chosen with `gptr_return()`, the turn count.

s$usage; s$cost; s$envir
Usage, total cost and the agent's workspace.

col = "mpg"; n = 3
peter("Plot {col} for the top {n} cars.")
Literal prompts interpolate {name} from R.

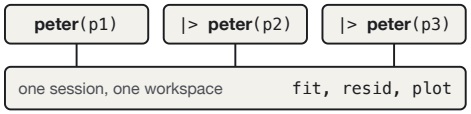
peter("Clean the data.", raw, mode = auto, model = "openai/gpt-6-sol")
Name other arguments exactly; bare names work.

peter("Summarise.", sales, budget = list(tokens = 5e4, turns = 5))
Per-call limits on tokens, cost and turns.

gptr_return(coef(fit)) # in agent R code
Marks the run's result, read back as s$value.

tryCatch(**peter**("Fit.", d), gptr_error = function(e) e$session)
A run that stops early carries its session as e$session.

Pipe: one session, many turns



s |> **peter**("Now plot the residuals.")
Same session and workspace, one more turn.

peter("Draft a summary.", fit) |> **peter**("Shorten it.")
Each pipe adds a turn to the session on its left.

b = **gptr_fork**(s)
b |> **peter**("Try a log model instead.")
Branch off; the original stays unchanged.

s |> **peter**("Add a caveat.", model = opus)
Switch the model mid-pipe; the history is kept.

s; **summary**(s); **gptr_last**()
Print it, its history, the most recent session.

The console

session = **peter**()
session |> **peter**()
Open the console; /exit returns the session.

AT THE peter> PROMPT

Fit mpg on wt.	A prompt for the session
!head(d)	Run R and show the model
!!x = 2	The same, with no note
```r … ```	Fenced R, run like !code
""" … """	A multi-line prompt
line \	Continue on the next line
@notes.md	Attach a file's first 40 lines
@fit	Attach an object
{n}	Insert an R value

SLASH COMMANDS

/help [cmd]	Commands and syntax
/model [ref]	Show or switch model
/mode [mode]	Show or switch mode
/status	Session, model, mode
/cost /context	Tokens and cost
/env [pattern]	Workspace objects
/tools	Tools and helpers
/compact [focus]	Compact the history
/clear	New chat, same objects
/resume [id]	Resume a stored session
/fork	Fork and continue there
/doc [path]	The history document
/skills /skill:name	List or use skills
/mcp	List MCP servers
/permissions	Show or change rules
/undo /redo /retry	Undo, redo, resend
/rewind [k]	Keep turns 1 to k
/exit	Leave; return the session

WHILE AN ANSWER RUNS

In a terminal (not Windows), type to steer. Ctrl-C pauses: **[s]**teer, **[f]**ollow-up, **[c]**ontinue, **[a]**bort, **[b]**ackground; in RStudio and Positron, Esc aborts. Approvals: **[y]**es, **[a]**lways, **[n]**o, **[?]**.

System 1 and System 2

System 1 typed decision → `TRUE "lung" 1.88` vector + probabilities

System 2 agent with tools → `gptr_session` $text, $value, objects

The model decides what `peter()` returns. A decision model (Jev, Clef) answers once per input; a conversational model works in turns.

SYSTEM 1: TYPED DECISIONS

rct = **peter**("Is this an RCT?", abstracts, model = jev)
One logical answer per element, TRUE when P(yes) ≥ threshold (0.5).

gptr_prob(rct)
P(yes) per element; "probabilities" gives the matrix.

peter("Which tissue?", cells, model = jev, choices = c("liver", "lung", "other"))
One label per input: a character vector.

peter("How severe?", events, model = jev, levels = c("low", "mid", "high"))
A score: the expected zero-based level, such as 1.88.

peter(q, x, model = jev, min_confidence = 0.8, uncertain = "stop")
Abstain below a confidence: NA by default, or a fallback or an error.

if (isTRUE(ok)) **message**("Escalate.")
abstracts[rct]; **sum**(rct)
Answers are plain vectors for if, for, subsetting and ifelse().

s |> **peter**("Did it succeed?", model = jev)
Judge a session's last answer; adds no turn.

m = if (isTRUE(hard)) "opus" else "haiku"
peter(task, model = m)
Route each task to a deep or a fast agent.

SYSTEM 2: AGENTS

peter("Plan it.", d, model = opus, .opts = list(thinking = "high"))
More reasoning: off to max; or a suffix, "anthropic/claude-sonnet-5-5:high".

Models and providers

```
"anthropic/claude-sonnet-5-5:high"
```
provider · model id · thinking

Aliases work too: `sonnet`, `opus`, `haiku`, `gpt`, `flash`, `jev`, `claude_code`, `codex`.

`anthropic/…`	Anthropic `ANTHROPIC_API_KEY`
`openai/…`	OpenAI `OPENAI_API_KEY`
`google/…`	Google Gemini `GEMINI_API_KEY`
`openrouter/…`	OpenRouter; also groq, deepseek, mistral, together, xai, cerebras and fireworks routes `<NAME>_API_KEY`
`azure/…`	Azure OpenAI `AZURE_OPENAI_API_KEY`, `AZURE_OPENAI_ENDPOINT`
`bedrock/…`	AWS Bedrock `AWS_BEARER_TOKEN_BEDROCK`, `AWS_REGION`
`ollama/…`	Ollama, local, no key; also lmstudio, llamacpp, vllm
`claude-cli`	Claude Code CLI `claude auth login`
`codex`	Codex CLI `codex login`
`jev`	System 1 · TypeSafe Jev `TYPESAFE_API_KEY`
`ollama/clef`	System 1 · local Clef and Clef Flash, Ollama 0.35.1 or later

gptr_models(provider = "ollama",
  refresh = TRUE)
Ask Ollama which models are installed; for a cloud route such as openrouter, download its catalog.

```
m = list(id = "my-id", tool_call = TRUE)
gptr_config(providers = list(
  bedrock = list(models = list(m))))
```
Declare Bedrock or Azure models, then `model = "bedrock/my-id"`. Each call replaces `providers`.

gptr_register(**gptr_provider**("corp",
  "openai-completions", base_url = u,
  auth = "CORP_KEY", models = list(m)))
Any OpenAI-compatible endpoint.
`api = "bedrock-converse"` signs Bedrock with SigV4.

```
eg = gptr_config()$egress
eg$openai = "ack"
gptr_config(egress = eg, .scope = "user")
```
Acknowledge sending context to a remote provider, for unattended runs.

The script is the history

Bind a document, and each top-level `peter()` call gets a block with the R code the agent ran.

```
peter("Fit mpg on wt in mtcars.")
# >>> gptr:46d369 model=…
fit = lm(mpg ~ wt, mtcars)
coef(fit)[["wt"]]
#> [1] -5.344472
## Decision: linear fit
# <<< gptr:46d369
```
prompt · header · code · output · note · end

gptr_doc("analysis.R")
Record calls into `.R`, `.Rmd`, `.qmd` or `.ipynb`.

gptr_source("analysis.R",
  replay = "replay")
Rerun: fresh blocks replay with no model call.

```
GPTR_REPLAY=replay Rscript analysis.R
```
Prove from the shell that a script needs no model.

gptr_blocks("analysis.R")
Each block's status: fresh, stale (prompt edited), user-edited or undone.

gptr_doc("report.Rmd")
rmarkdown::render("report.Rmd")
Render as usual; each block becomes a chunk named `gptr-<id>`.

gptr_doc("analysis.ipynb", sync = TRUE)
Apply blocks queued while it was open.

gptr_doc("log.R", format = "transcript")
Record the console as a runnable `.R` transcript.

peter(q, replay = "live")
Modes: `"auto"` (default), `"replay"`, `"live"`, `"record"`.

EXPORT A RECORDED .R

Export needs `peter()` calls with literal arguments.

gptr_export("analysis.R", "plain.R")
Plain R: recorded code runs, `peter()` calls become comments.

gptr_export("analysis.R", "peter.R",
  format = "peter")
A `peter()` script: calls run, recorded code is commented.

Try it offline

A scripted model runs every pattern with no key, server or network.

```
fake = gptr_fake_provider(list(
  list(tool = "r", input = list(
    code = "fit = lm(mpg ~ wt, mtcars)")),
  "Each 1000 lb lowers mpg by about 5.3."))
s = peter("Fit mpg on wt.", mtcars,
  model = fake, mode = auto)
s$text; coef(fit)
```
The first reply runs R; the second is the answer.

Tools, files and artifacts

The model calls `read`, `r`, `edit`, `write` and `ask`. Inside `r` it uses the `peter$` helpers, which you can call too.

peter("Review R/.", tools = extended)
Presets: `minimal`, `standard`, `readonly`, `extended` (adds grep, find, ls).

peter(q, tools = c("-write", "+grep"))
Remove or add single tools.

peter("What is in data/scores.csv?")
The read tool returns text and images; binary files get a loading hint.

peter$read("R/fit.R", limit = 40)
peter$grep("lm\\(", path = "R")
peter$find**("*.R"); **peter$ls("R")
Read a file window, search contents, find and list files.

peter$edit("R/fit.R", list(list(
  oldText = "n = 10", newText = "n = 20")))
peter$write("notes.txt", "done\n")
Exact replacements; atomic writes.

peter$sh(c("git", "status"))
peter$bg(c("Rscript", "long.R"))
Run a program; run one in the background.

peter$sql("SELECT cyl, AVG(mpg) AS mpg
  FROM mtcars GROUP BY cyl", mtcars)
SQL on data frames (duckdb);
`peter$py()` runs Python.

ARTIFACTS: SHINY APPS

peter("Make a Shiny app of it.", survey)
The agent writes `.gptr/artifacts/<id>/app.R` and launches it.

gptr_artifacts()
gptr_artifacts("survey-view", open = TRUE)
List versioned apps; open, stop or relaunch a version.

Permissions and trust

`plan`	Read only; R runs in a scratch environment
`manual`	Default: read-only actions run, others ask
`edits`	Also edit and write files in the project
`auto`	All but critical actions, gptr settings and secret reads
`yolo`	Nearly all; critical actions still ask

gptr_permissions(deny = "r(sh:rm *)")
Rules: `deny` beats `ask` beats `allow`.

gptr_rewind(s)
Undo the last turn: objects, files and conversation.

The gate is advisory, not a sandbox;
see `?gptr_security`.

Skills, plugins and MCP

gptr_skills()
Skills from the project, user, plugins and packages.

peter("Speed up f.", f,
  skills = "high-performance-r")
Preload a skill; yours go in `.gptr/skills/<name>/SKILL.md`.

`.gptr/prompts/check.md  # becomes /check`
A prompt template is a console command.

peter("Report units.", plugins = "./lab")
Enable a plugin; `gptr_plugins()` lists them.

gptr_mcp_add("docs",
  url = "http://127.0.0.1:8000/mcp")
Add an MCP server (or `command =` for stdio).

petermcp<server>$<tool>(...)
MCP tools are R functions the agent calls.

```
h = gptr_mcp_serve(); h$stop()
```
Serve this R session to Claude Code, Codex or Cursor.

```
up = gptr_tool("shout", "Upper-case text",
  fun = function(x) toupper(x))
off = gptr_register(up)
```
Add your own tool; `off()` removes it.

Teams, fan-outs, background

```
s = peter("Review fit.", agents = list(
  stats = agent(system = "Check stats."),
  code = agent(system = "Check code.")))
```
A team of named roles; `s$stats$text` reads one.

```
f = peter("Summarise.", cohorts,
  parallel = 2)
```
One child per element, two at a time; each returns with `gptr_return()`.

```
gptr_parallel(a = peter("Plan."),
  b = peter("Review."))
```
Run independent calls concurrently.

```
b = peter("Check fit.", fit,
  background = TRUE)
gptr_wait(b)
```
Experimental: return at once; also `gptr_steer()`, `gptr_cancel()`, `gptr_jobs()`.

gptr_usage(s, by = "agent")
Tokens and cost by session, agent, model or route.

Learn more

browseVignettes("gptr")
All guides; start with getting-started.